UNIVERSITY OF UYO

DEPARTMENT OF METHEMATICS

MTH211 TEST Mathematical Method 1 2025/2026 SESSION

Instruction: Answer all the questions Time: 30 minutes

1. State the hypothesis of the mean value theorem and hence find the value of $c \in [-2,3]$ for which $f(x) = 3x^3 - x^2 + 5x - 4$ satisfies the theorem.
2. Prove that $\lim_{(x,y)\to(1,2)} x^2 + 2y = 5$ and in particular, find δ if $\varepsilon = 0.0025$
3. Find the Taylor's series expansion of the of the function , $f(x) = \ln(1 + x)$ about the origin up to the cubic term. Hence find an approximation to $\ln(1.5)$.

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